Port Security – Practical Lab Documentation

Lab Completed

CCNA / NOC Practice

1. Lab Objective

The objective of this lab was to understand and implement Cisco Switch Port Security to prevent unauthorized devices from connecting to a switch port.

We learned how to:

- Enable Port Security on a switch.
- Configure maximum allowed MAC addresses.
- Enable Sticky MAC learning.
- Generate a security violation.
- Identify unauthorized devices.
- Understand the Shutdown violation mode.
- Troubleshoot and recover an err-disabled port.
- Understand the difference between IP addresses and MAC addresses.
- Understand MAC address spoofing.

2. Devices and Tools Used

We used Cisco Packet Tracer to create and configure the network topology.

Device	Quantity	Purpose
Cisco 2960 Switch	1	Implement Port Security
PC 1	1	Authorized device
PC 2	1	Unauthorized device
Copper Straight-Through Cable	2	Connect PCs to switch

Total devices: 3

3. Network Topology

How to configure Switch Port Security in Packet Tracer – Computer Networking Tips

computernetworking747640215.wordpress.com

We connected both PCs to the same switch, but tested them one at a time using the same switch port.

Device	Switch Port	IP Address
PC 1	Fa0/1	192.168.1.2
PC 2	Fa0/1 (one at a time)	192.168.1.2

Important: Both PCs used the same IP address during separate tests. However, their MAC addresses were different.

4. Initial Configuration

First, we connected PC 1 to Switch Fa0/1.

We configured the PC with:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Then we verified connectivity by pinging the destination device at 192.168.1.2 in our lab.

5. Enable Port Security

We entered the following commands on the switch:

enable

configure terminal

interface fastethernet 0/1

switchport mode access

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation shutdown

end

Explanation of Commands

Command	Purpose
enable	Enter privileged EXEC mode
configure terminal	Enter global configuration mode
interface fa0/1	Select interface Fa0/1
switchport mode access	Configure the port as an access port
switchport port-security	Enable Port Security
maximum 1	Allow only one secure MAC address
mac-address sticky	Automatically learn and save MAC addresses
violation shutdown	Shut down the port when a violation occurs
end	Exit configuration mode

6. Sticky MAC Address Learning

After configuring Port Security, we generated traffic from PC 1.

The switch automatically learned its MAC address.

We checked it using:

show port-security address

Our output showed:

Vlan Mac Address Type Ports

1 000B.BE43.D110 SecureSticky Fa0/1

What happened?

The switch learned PC 1's MAC address and added it to the running configuration through Sticky MAC learning.

We also checked:

show port-security

Initially, the output showed:

MaxSecureAddr: 1

CurrentAddr: 1

SecurityViolation: 0

This confirmed that one MAC address was learned successfully.

7. Verify Port Security Configuration

We used:

show port-security interface fa0/1

Our important results were:

Parameter	Result
Port Security	Enabled
Port Status	Secure-up
Violation Mode	Shutdown
Maximum MAC Addresses	1
Total MAC Addresses	1
Configured MAC Addresses	0
Sticky MAC Addresses	1
Security Violation Count	0

This confirmed that Port Security was working correctly.

8. Generate a Security Violation

Now we disconnected PC 1 and connected PC 2 to the same switch port, Fa0/1.

PC 2 had the same IP address:

192.168.1.2

But its MAC address was different.

PC	MAC Address
PC 1	000B.BE43.D110

PC 2 00E0.B005.9C26

The switch detected the new MAC address and generated a security violation.

Our switch displayed:

%PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/1

%PORT_SECURITY-2-PSECURE_VIOLATION:

Security violation occurred, caused by MAC address 00E0.B005.9C26

The switch automatically placed Fa0/1 into an err-disabled state.

Verification

show port-security interface fa0/1

Important output:

Port Status : Secure-shutdown

Violation Mode : Shutdown

Maximum MAC Addresses : 1

Sticky MAC Addresses : 1

Security Violation Count : 1

Security violation successfully generated

9. Why Was PC 2 Blocked Even Though It Had the Same IP Address?

This was one of the most important concepts we learned.

Port Security checks MAC addresses, not IP addresses.

- PC 1 had MAC address 000B.BE43.D110.
- PC 2 had MAC address 00E0.B005.9C26.

The switch had already saved PC 1's MAC address through Sticky MAC learning.

Therefore, PC 2 was considered an unauthorized device.

Remember: Two devices can have the same IP address, but different MAC addresses. Port Security uses MAC addresses to identify authorized devices.

10. Port Recovery

After a security violation, simply reconnecting the original PC did not restore connectivity.

The switch port remained in an err-disabled state.

We needed to manually recover the port.

Commands:

enable

configure terminal

interface fa0/1

shutdown

no shutdown

end

Then we verified the interface:

show port-security interface fa0/1

Expected status after recovery, with the original PC connected:

Port Status : Secure-up

Note: The violation counter can remain at 1 after recovery. This is normal.

11. MAC Address Spoofing

We also learned that MAC addresses are generally assigned by the network adapter manufacturer.

However, an operating system or supported network adapter can temporarily use a different MAC address through MAC spoofing.

For example:

- Original MAC: 00E0.B005.9C26
- Spoofed MAC: 000B.BE43.D110

If another device spoofs an authorized MAC address, basic Port Security may not distinguish it from the original device.

This is why Port Security alone is not a complete network security solution.

12. Important Verification Commands

Command	Purpose
show port-security	Display Port Security status
show port-security address	Display secure MAC address table
show port-security interface fa0/1	Display detailed port security information
show running-config	Display active configuration
show interfaces fa0/1	Check interface status
show mac address-table	Display learned MAC addresses

Interview Questions and Answers

Here are 15 interview questions based on the practical work we performed.

Q1. What is Port Security?

Port Security is a Cisco switch feature that controls which MAC addresses are allowed to access a switch port.

Q2. Why do we use Port Security?

We use Port Security to prevent unauthorized devices from connecting to the network through switch ports.

Q3. What is a MAC address?

A MAC address is a hardware-associated address used to identify a network interface at Layer 2 of the OSI model.

Q4. What is Sticky MAC learning?

Sticky MAC learning automatically learns MAC addresses and adds them to the running configuration.

Q5. What is the maximum MAC address limit?

It defines how many secure MAC addresses are allowed on a particular switch port.

Q6. What happens when an unauthorized device connects?

The switch detects a security violation. The action depends on the configured violation mode.

Q7. What is an err-disabled port?

An err-disabled port is a switch port automatically disabled because of an error condition or security violation.

Q8. Why did our Fa0/1 port go into err-disabled state?

Because we connected PC 2 with a different MAC address while the port was configured to allow only one secure MAC address.

Q9. How did we recover the port?

We used the shutdown and no shutdown commands under interface configuration mode.

Q10. What is the difference between IP and MAC addresses?

An IP address is a logical Layer 3 address, while a MAC address is a Layer 2 address associated with a network interface.

Q11. Can two PCs have the same IP address?

Yes, they can be manually configured with the same IP, but simultaneous use in the same network can cause an IP address conflict.

Q12. Can a MAC address be changed?

Yes. It can be temporarily changed through MAC spoofing if supported by the operating system and network adapter.

Q13. What is the difference between Secure-up and Secure-shutdown?

Secure-up means the secured port is operational. Secure-shutdown means the port has been disabled because of a security violation.

Q14. Does Port Security check IP addresses?

No. Traditional Cisco Layer 2 Port Security checks MAC addresses, not IP addresses.

Q15. What is the difference between Sticky MAC and Static MAC?

- Sticky MAC automatically learns MAC addresses.
- Static MAC requires the administrator to manually configure the MAC address.

13. What We Learned

Learning Outcomes

1. Basic Cisco switch configuration.

2. Access port configuration.
3. Port Security implementation.
4. Maximum MAC address configuration.
5. Sticky MAC address learning.
6. MAC address table verification.
7. Security violation detection.
8. Err-disabled port troubleshooting.
9. Port recovery using shutdown and no shutdown.
10. IP address vs MAC address concepts.
11. MAC spoofing fundamentals.

14. GitHub Repository Details

You can use these details when uploading your lab:

Repository name:

Cisco-Port-Security-Lab

Description:

Cisco Switch Port Security practical lab using Cisco Packet Tracer. Includes Sticky MAC learning, MAC address limitation, security violation testing, err-disabled troubleshooting, and port recovery.

Skills demonstrated:

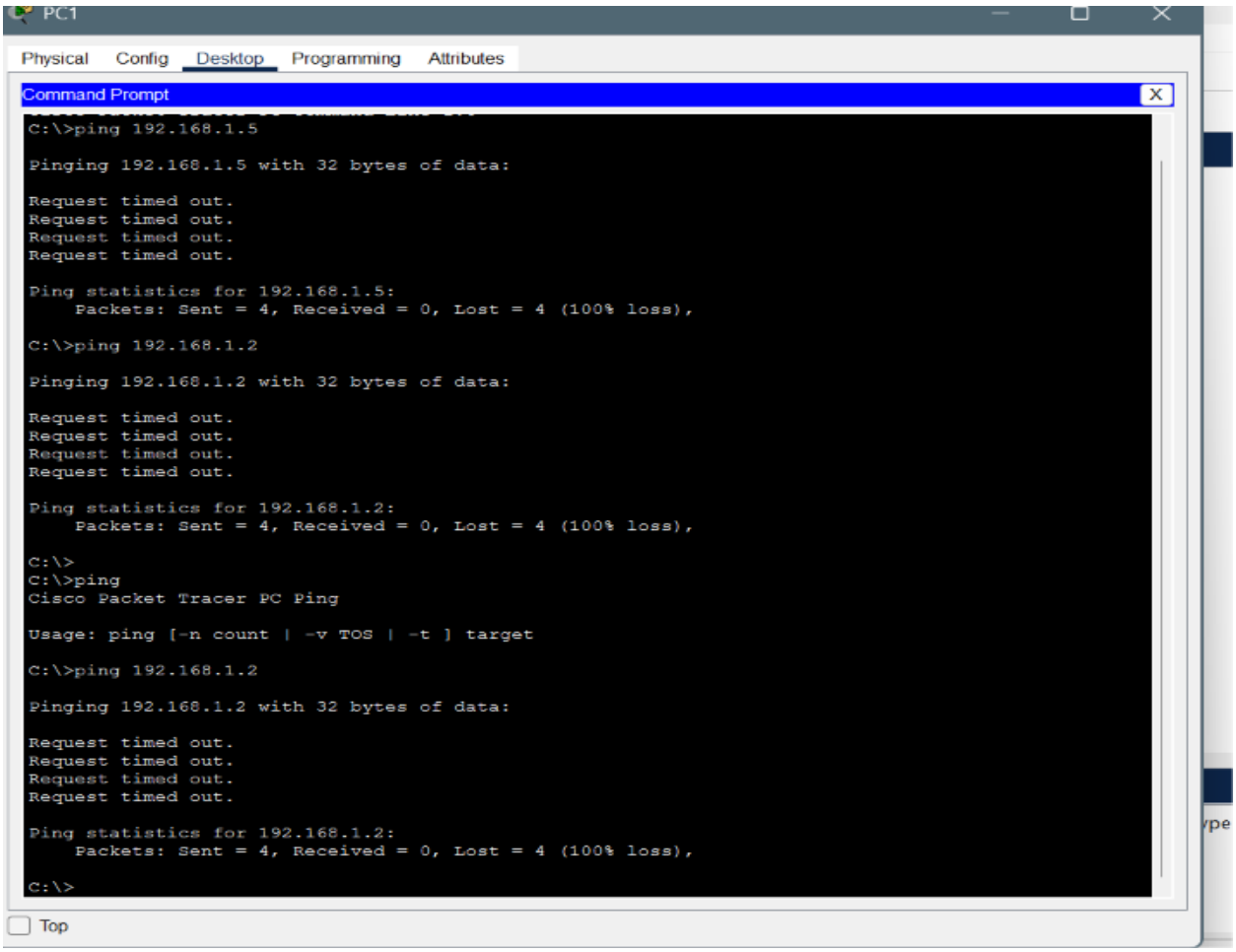
Cisco IOS, Switching, Port Security, Sticky MAC, MAC Address Learning, Network Troubleshooting, CCNA

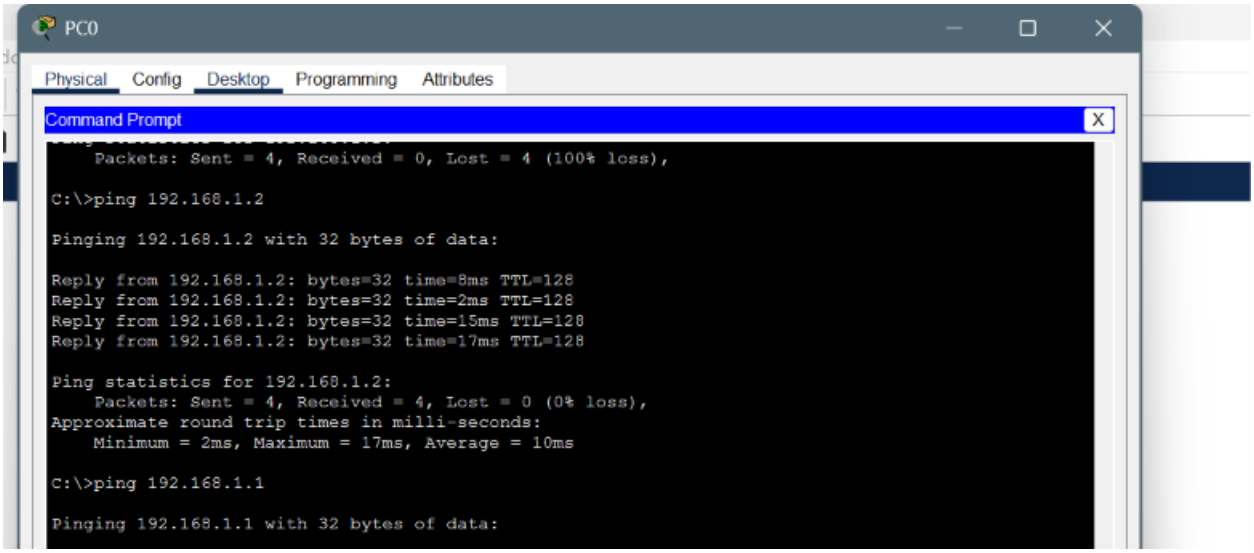
15. Next Practical Tasks

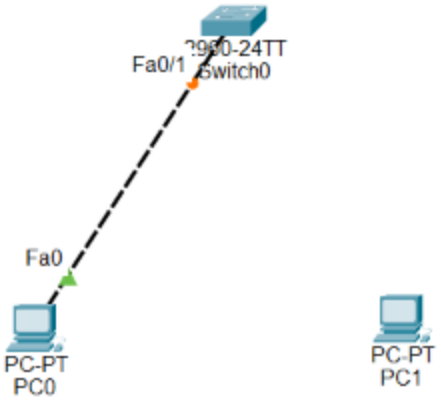
You have completed the basic Shutdown-mode test. These are the remaining practical exercises to make your lab more complete.

No.	Practical task	Status
1	Enable Port Security	Done

2	Configure Maximum MAC = 1	Done
3	Configure Sticky MAC	Done
4	Generate security violation	Done
5	Recover err-disabled port	Practice / verify
6	Test Protect mode	Pending
7	Test Restrict mode	Pending
8	Compare all three violation modes	Pending


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.1.5
Pinging 192.168.1.5 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.5:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
C:\>ping
Cisco Packet Tracer PC Ping
Usage: ping [-n count | -v TOS | -t] target
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=8ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=15ms TTL=128
Reply from 192.168.1.2: bytes=32 time=17ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 17ms, Average = 10ms
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:



I use one pc for better uderstandig and simple or easy configuration it's just practis

